DYNAMIC PHYSICAL PROFILING IN FOOTBALL

Project overview | Work in progress

Current stage: data preparation, quality control, analytical filtering, and raw player-level physical profiling.

R | tidyverse | ggplot2 | GPS | Heart rate | RPE

1. Project purpose

The aim of this project is to build a reproducible workflow for describing the physical profile of football players from repeated training-session data. Rather than treating the final radar chart as the project itself, the profile is the final product of a full analysis pipeline: preprocessing, data-quality assessment, analytical inclusion, player-level aggregation, and comparison with meaningful reference groups.

The current version focuses on descriptive profiling. External-load metrics derived from GPS data are presented separately from internal-response indicators derived from perceived exertion and heart-rate information.

2. Data source

The project uses the open dataset published by Nijland, Toering, Ivarsson, de Jong and Lemmink (2025), "Dataset belonging to Relationship Between Academic Stress, Mental Fatigue and Training Load in Adolescent Soccer Players", DataverseNL.

The source study monitored 35 adolescent football players during the second half of the 2022-2023 season. The available material includes Polar Team Pro GPS and heart-rate data, RPE, training context, player position, and supporting variables. The original dataset is licensed under CC-BY-SA.

3. Unit of analysis

The analytical dataset is organised at player-session level: one row represents one player in one training session. Repeated sessions can then be summarised to construct a more stable description of each player's typical training demands.

4. Current workflow

Raw combined data - Player-session GPS, HR, RPE and contextual variables.

Preprocessing - Join player position and session type, harmonise duration information, and derive GPS/HR quality variables.

Relative load metrics - Express external-load measures relative to the actual available GPS duration.

EDA and quality control - Inspect missingness, session duration, recording quality, distributions, correlations, contextual differences and suspicious sessions.

Analytical filtering - Retain valid sessions and players with enough repeated observations.

Player-level summaries - Use session medians to describe each player's typical physical demands.

Reference percentiles - Compare each player with the full sample and with players in the same position.

Physical profile - Combine external-load percentiles and internal-response distributions in a player card.

5. External-load profile

The preprocessing stage creates relative external-load variables using the actual available GPS duration, improving comparability across sessions of different lengths.

- Distance per minute
- Low-speed activity (LSA) per minute
- Moderate-speed running (MSR) per minute
- High-speed running (HSR) per minute
- Sprint distance per minute
- Combined HSR + sprint distance per minute
- Number of sprints per minute
- Accelerations per minute
- Decelerations per minute

In the current raw profile, the player's typical value for each metric is the median across eligible sessions. These medians are converted to percentile ranks for overall-sample and position-specific comparisons.

6. Internal-response profile

Internal response is kept separate from external load. The current player card summarises:

- RPE (rating of perceived exertion)
- TRIMP per minute
- Percentage of recorded HR duration spent in HR zones 4 and 5

These variables are displayed against their reference distributions rather than merged into the external-load radar. This preserves the distinction between the work performed and the player's physiological or perceptual response.

7. Data quality and analytical inclusion

Before player profiles are created, the descriptive-analysis script evaluates missingness, session duration, GPS/HR recording quality, external-load distributions, contextual effects and potentially suspicious observations.

- External-load data must be available.
- GPS duration must be at least 10 minutes.
- GPS missingness must be no greater than 10% when the missingness estimate is available.
- Players must have at least 20 valid sessions to enter the final analytical dataset.

Potential extreme values are flagged for review rather than automatically removed, because genuinely demanding football sessions may generate extreme physical-load values.

8. Current output

The main output is a player card with two complementary components. The external-load radar displays percentile positioning across physical-load metrics, while the internal-response panel locates the player's value within the corresponding reference distribution.

The card also reports the number of sessions supporting the external, HR and RPE profiles and assigns a simple evidence category based on data availability. For position-specific comparisons, the number of players in the reference position is also reported.

9. What the project currently demonstrates

- Construction of an analysis-ready football training-load dataset from multiple data sources.
- Practical handling of GPS and HR recording-quality information.
- Exploratory analysis of missingness, distributions, correlations and contextual effects.
- Transparent analytical inclusion rules before player comparison.
- Conversion of repeated player-session observations into interpretable player-level summaries.
- Overall and position-specific benchmarking.
- Reproducible visual reporting in R.

10. Project status and next development

> **Work in progress:** The scripts currently document the descriptive/raw-profile stage. A later version can add context-adjusted modelling so that player estimates are separated from systematic session-context effects. That modelling stage is not yet part of the completed workflow described here.

11. Repository structure

```
dynamic-physical-profiling/
 README.md
 code/
  00_preprocessing.R
  01_descriptive_analysis.R
  02_raw_profile.R
 data/
 data_source/
  original_dataset_README.pdf
 figures/
```

12. Attribution

Nijland, Rick; Toering, Tynke; Ivarsson, Andreas; de Jong, Johan; Lemmink, Koen (2025). "Dataset belonging to Relationship Between Academic Stress, Mental Fatigue and Training Load in Adolescent Soccer Players" [Data set]. DataverseNL. License: CC-BY-SA.